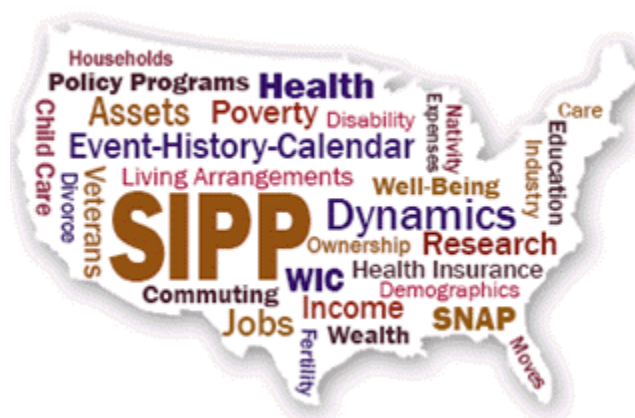


Survey of Income and Program Participation 2018+ Data Primer



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Introduction

This guide is intended to be a primer for working with 2018+ Survey of Income and Program Participation (SIPP) data. The [SIPP Users' Guides](#) provide comprehensive information about the survey and its data, available on the [SIPP website](#). Other important information is below.

- SIPP data are available in STATA, SAS, and pipe-delimited text format on the [SIPP Datasets pages](#)
- The [Online Codebook](#) and [Data Dictionaries](#) provide metadata for all variables, including those discussed in this guide
- [SIPP User Notes](#) provide nuances, notes, and data issues that users should be aware of
- The [SIPP Library](#) contains publications, reports, and other data products published by the Census Bureau
- In SIPP data, a *missing* value for a variable, often labeled by “.”, indicates the record was not eligible to receive a value as defined by the variable’s “universe”

About the Data in This Guide

The data in this guide are fabricated for demonstration purposes. Matching **SSUIDs** can be found in the 2022 SIPP public-use file, but all adjoining data were created for illustration purposes only, and any similarities between these data and the public-use data is purely coincidental.

Background

Each year, Census Bureau field representatives (FRs) interview tens of thousands of households across the country who are sampled for the SIPP. The SIPP is a longitudinal survey, so these households are interviewed for multiple years. For simplicity, this guide starts with households that are being interviewed for the first time (their “Wave 1” interview). Later sections will discuss respondents moving to different households, as the SIPP follows household members who relocate after their initial interview.

All sampled households are assigned a randomized “Sample Unit Identifier”, or SSUID.

SSUID	
	10111110312318
	00013364117220
	20037909398822
	00011428546521
	03041809214921
	50081893237320
<i>(and so on...)</i>	

Within each household, each person assigned a “Person Number”, or PNUM.

SSUID		PNUM	
	10111110312318		101
	10111110312318		102
	10111110312318		103
	10111110312318		104
	00013364117220		101
	00013364117220		102
	00013364117220		103
<i>(and so on...)</i>			

Of note, the last two digits of an **SSUID** identify which *Panel* the household is a part of (i.e., which year they were first sampled in – 2018, 2019, 2020, etc.). The first digit of **PNUM** identifies which *Wave* a person entered the **SSUID** (i.e. which number interview they joined the household – 1st, 2nd, 3rd, etc.).

Building the Data

SIPP interviews usually occur in February through May. Most questions ask about the previous calendar year, known as the “reference year”. Some questions also measure up to – and include – the “time of interview”.

Reference Year												Interview Year			
January	February	March	April	May	June	July	August	September	October	November	December	January	February	March	
←															→
															

In SIPP data, the 12 months of the reference year (January through December of the calendar year preceding the interview) are translated into 12 rows for each respondent^{1,2}. This is known as a “person-month- level” microdata structure: each row of the dataset represents one month for one interviewed person.

Row	SSUID	PNUM	MONTHCODE	(other data...→)
1	10111110312318	101	1	...
2	10111110312318	101	2	...
3	10111110312318	101	3	...
4	10111110312318	101	4	...
5	10111110312318	101	5	...
6	10111110312318	101	6	...
7	10111110312318	101	7	...
8	10111110312318	101	8	...
9	10111110312318	101	9	...
10	10111110312318	101	10	...
11	10111110312318	101	11	...
12	10111110312318	101	12	...
13	10111110312318	102	1	...
14	10111110312318	102	2	...
15	10111110312318	102	3	...
16	10111110312318	102	4	...
17	10111110312318	102	5	...
18	10111110312318	102	6	...
(and so on...)				

The **MONTHCODE** variable represent which month a row of data refers to. Putting this together with sample ID and person ID, unique records in SIPP are defined by a combination of **SSUID + PNUM + MONTHCODE**. Through this unique data structure, you can easily measure month-to-month changes.

¹ Some SIPP questions include information about the interview year, but published data only include “time of interview” and “reference year” data.

² In rare cases, respondents have fewer than 12 monthly records (such as for newborns and persons temporarily being “out of sample” during the reference year).

Person-Month-Level Data

Let's add some data for *Person 101* and *Person 102* in **SSUID 10111110312318**: **TPTOTINC** ("*Total personal income for the month*").

Row	SSUID	PNUM	MONTHCODE	TPTOTINC
1	10111110312318	101	1	2354
2	10111110312318	101	2	2204
3	10111110312318	101	3	2140
4	10111110312318	101	4	3400
5	10111110312318	101	5	1956
6	10111110312318	101	6	2354
7	10111110312318	101	7	2204
8	10111110312318	101	8	2140
9	10111110312318	101	9	3400
10	10111110312318	101	10	1956
11	10111110312318	101	11	2204
12	10111110312318	101	12	3405
13	10111110312318	102	1	2054
14	10111110312318	102	2	1904
15	10111110312318	102	3	1840
16	10111110312318	102	4	3100
17	10111110312318	102	5	1656
18	10111110312318	102	6	2054
19	10111110312318	102	7	2491
20	10111110312318	102	8	2418
21	10111110312318	102	9	3842
22	10111110312318	102	10	2210
23	10111110312318	102	11	2491
24	10111110312318	102	12	3848

(and so on...)

The variable **TPTOTINC** describes income for this person for each month of the reference year. This is referred to as "person-month" data because its value can change from month to month. Person-month data are the most straightforward type of SIPP data. Other SIPP data types are discussed below.

Spell-Level Data

Not all SIPP data vary from month to month. "Spell-level" data are an example of this. In the SIPP, the term *spell* is used to describe a continuous period of time a respondent received, participated in, was covered by, or was in a state of a given topic. Let's use **EJB1_BMONTH** and **EJB1_EMONTH** as an example (the first and last months of a job spell for *Job #1*, respectively).

Row	SSUID	PNUM	MONTHCODE	TPTOTINC	EDISABL	EJB1_BMONTH	EJB1_EMONTH
1	10111110312318	101	1	2354	2	1	4
2	10111110312318	101	2	2204	2	1	4
3	10111110312318	101	3	2140	2	1	4
4	10111110312318	101	4	3400	2	1	4
5	10111110312318	101	5	1956	2	.	.
6	10111110312318	101	6	2354	2	.	.
7	10111110312318	101	7	2204	2	.	.
8	10111110312318	101	8	2140	2	8	11
9	10111110312318	101	9	3400	2	8	11
10	10111110312318	101	10	1956	2	8	11
11	10111110312318	101	11	2204	2	8	11
12	10111110312318	101	12	3405	2	.	.
13	10111110312318	102	1	2054	1	.	.
14	10111110312318	102	2	1904	1	.	.
15	10111110312318	102	3	1840	1	3	10
16	10111110312318	102	4	3100	1	3	10
17	10111110312318	102	5	1656	1	3	10
18	10111110312318	102	6	2054	1	3	10
19	10111110312318	102	7	2491	1	3	10
20	10111110312318	102	8	2418	1	3	10
21	10111110312318	102	9	3842	1	3	10
22	10111110312318	102	10	2210	1	3	10
23	10111110312318	102	11	2491	1	.	.
24	10111110312318	102	12	3848	1	.	.

(and so on...)

Spell-level data are copied to all months of a "spell" and are missing during the months that no spell occurred. In the above data, *Person 101* had two spells of work at their *Job #1*: January through April, and August through November. *Person 102* had one spell of work at their *Job #1*: March through October. For programming spell-level analysis, we recommend subsetting the data to one record per spell with something like "unique(SSUID, PNUM, EJB1_BMONTH)" or "where EJB1_BMONTH = MONTHCODE" where appropriate to avoid including duplicate records. This results in a spell-level subset of data like the below table.

Row	SSUID	PNUM	EJB1_BMONTH	EJB1_EMONTH
1	10111110312318	101	1	4
2	10111110312318	101	8	11
3	10111110312318	102	3	10
4	00013364117220	101	1	5
5	00013364117220	101	9	12

(and so on...)

Person-Level Data

Another type of SIPP data that do not vary from month to month are called “person-level” data. These data are generally based on questions about the entire reference year or a person’s current situation or status at the time of their interview. The below table uses **EDISABL** (“Do you have a physical, mental, or other health condition that limits the kind or amount of work you can do?”) as an example.

Row	SSUID	PNUM	MONTHCODE	TPTOTINC	EDISABL
1	10111110312318	101	1	2354	2
2	10111110312318	101	2	2204	2
3	10111110312318	101	3	2140	2
4	10111110312318	101	4	3400	2
5	10111110312318	101	5	1956	2
6	10111110312318	101	6	2354	2
7	10111110312318	101	7	2204	2
8	10111110312318	101	8	2140	2
9	10111110312318	101	9	3400	2
10	10111110312318	101	10	1956	2
11	10111110312318	101	11	2204	2
12	10111110312318	101	12	3405	2
13	10111110312318	102	1	2054	1
14	10111110312318	102	2	1904	1
15	10111110312318	102	3	1840	1
16	10111110312318	102	4	3100	1
17	10111110312318	102	5	1656	1
18	10111110312318	102	6	2054	1
19	10111110312318	102	7	2491	1
20	10111110312318	102	8	2418	1
21	10111110312318	102	9	3842	1
22	10111110312318	102	10	2210	1
23	10111110312318	102	11	2491	1
24	10111110312318	102	12	3848	1

(and so on...)

“Person-level” data are copied across all monthly records for a person. When analyzing person-level data, you’ll likely only want to use one value per person. For programming, we recommend subsetting the data to one record per person with something like `unique(SSUID, PNUM, EDISABL)` or `where MONTHCODE = 12` (since month 12 is the record closest to the interview month) where appropriate. This results in a person-level subset of data like the below table.

Row	SSUID	PNUM	EDISABL
1	10111110312318	101	2
2	10111110312318	102	1
3	00013364117220	101	1
4	00013364117220	102	2
5	20037909398822	101	2

(and so on...)

Household-Level Data

The last major type of SIPP data are “household-level” data: data that are the same for all persons and months within a household. First, though, let’s discuss how to identify household members.

Row	SSUID	PNUM	SHHADID	MONTHCODE	(other data...→)
1	10111110312318	101	011	1	...
2	10111110312318	101	011	2	...
3	10111110312318	101	011	3	...
4	10111110312318	101	011	4	...
5	10111110312318	101	011	5	...
6	10111110312318	101	011	6	...
7	10111110312318	101	011	7	...
8	10111110312318	101	011	8	...
9	10111110312318	101	011	9	...
10	10111110312318	101	011	10	...
11	10111110312318	101	011	11	...
12	10111110312318	101	011	12	...
13	10111110312318	102	011	1	...
14	10111110312318	102	011	2	...
15	10111110312318	102	011	3	...
16	10111110312318	102	011	4	...
17	10111110312318	102	011	5	...
18	10111110312318	102	011	6	...

(and so on...)

Following the previous example, above is a household with two people - *Person 101* and *Person 102*. The variable **SHHADID** is an identifier (ID) that tells you which household people within the same **SSUID** lived in when they were interviewed. We can tell they’re in the same household because their **SSUIDs** and **SHHADIDs** match.³

You might be wondering, “*Why can’t we just use SSUID to identify household members?*” When someone moves out of the originally sampled household between interviews, SIPP follows them to their new household. Because they retain their original **SSUID**, we must use **SHHADID** to account for multiple households within an **SSUID**.

The below table brings in the variable **ERELRPE**, which tells us how everyone within a household is related.

Row	SSUID	PNUM	SHHADID	ERELRPE	MONTHCODE	(other data...→)
1	10111110312318	101	011	1	1	...
2	10111110312318	101	011	1	2	...
3	10111110312318	101	011	1	3	...
4	10111110312318	101	011	1	4	...
5	10111110312318	101	011	1	5	...
6	10111110312318	101	011	1	6	...
7	10111110312318	101	011	1	7	...
8	10111110312318	101	011	1	8	...

³ The variable **ERESIDENCEID** provides the monthly household ID for each person during the reference period.

Row	SSUID	PNUM	SHHADID	ERELRPE	MONTHCODE	(other data...→)
9	10111110312318	101	011	1	9	...
10	10111110312318	101	011	1	10	...
11	10111110312318	101	011	1	11	...
12	10111110312318	101	011	1	12	...
13	10111110312318	102	011	10	1	...
14	10111110312318	102	011	10	2	...
15	10111110312318	102	011	10	3	...
16	10111110312318	102	011	10	4	...
17	10111110312318	102	011	10	5	...
18	10111110312318	102	011	10	6	...
(and so on...)						

In the above table, *Person 101* is the reference person - the person whose name appears first on the deed or lease of the housing unit - and *Person 102* is their sibling (**ERELRPE** = 10). So now, what does household-level data look like?

Row	SSUID	PNUM	SHHADID	ERELRPE	MONTHCODE	EOWN_VEH
1	10111110312318	101	011	1	1	1
2	10111110312318	101	011	1	2	1
3	10111110312318	101	011	1	3	1
4	10111110312318	101	011	1	4	1
5	10111110312318	101	011	1	5	1
6	10111110312318	101	011	1	6	1
7	10111110312318	101	011	1	7	1
8	10111110312318	101	011	1	8	1
9	10111110312318	101	011	1	9	1
10	10111110312318	101	011	1	10	1
11	10111110312318	101	011	1	11	1
12	10111110312318	101	011	1	12	1
13	10111110312318	102	011	10	1	1
14	10111110312318	102	011	10	2	1
15	10111110312318	102	011	10	3	1
16	10111110312318	102	011	10	4	1
17	10111110312318	102	011	10	5	1
18	10111110312318	102	011	10	6	1
19	10111110312318	102	011	10	7	1
20	10111110312318	102	011	10	8	1
21	10111110312318	102	011	10	9	1
22	10111110312318	102	011	10	10	1
23	10111110312318	102	011	10	11	1
24	10111110312318	102	011	10	12	1
(and so on...)						

Household-level data are copied to all records of persons in the same household. Using the variable **EOWN_VEH** (“Whether any member of the household was the owner of a car, truck, or van during the reference period”) as an example in the above table, the value of “1” is copied to all records for both persons.

When analyzing household-level data, you'll likely only want to use only one value per household. Otherwise, you are including repeat values in your analysis. For programming, we recommend subsetting the data to one record per household with something like `"unique(SSUID, SHHADID, EOWN_VEH)"` or `"where MONTHCODE = 12 and ERELPRPE in (1,2)"`⁴ where appropriate. This results in a household-level subset of data like the below table.

Row	SSUID	SHHADID	EOWN_VEH
1	10111110312318	011	1
2	00013364117220	011	1
3	20037909398822	011	2
4	20037909398822	021	1
5	00011428546521	011	2
<i>(and so on...)</i>			

⁴ Subsetting with `"ERELPRPE in (1,2)"` implicitly pulls one value per **SHHADID** because each household has only one reference person.

Putting All Data Types Together

Below is an example table that puts the above data-level concepts together – person-month-, spell-, person-, and household-level data. Within one sample unit (**SSUID** 50081893237320), there are two households: **SHHADIDs** 011 and 021. Household 011 has two persons: **PNUMs** 101 and 203. Household 021 has one person: **PNUM** 102 who moved out of the originally sampled household. **TPTOTINC** is person-month-level data that varies by person by month. **EDISABL** is person-level data that only varies by person – and is copied across all months. **EJB1_BMONTH** and **EJB1_EMONTH** are spell-level data that only vary by spell – and are copied across all months of a spell. **EOWN_VEH** is household-level data that only varies by household – and is copied across all members of the household.

Row	SSUID	PNUM	SHHADID	MONTH CODE	TPTOTINC	EDISABL	EJB1_ BMONTH	EJB1_ EMONTH	EOWN_VEH
1	50081893237320	101	011	1	1177	2	1	12	1
2	50081893237320	101	011	2	1102	2	1	12	1
3	50081893237320	101	011	3	1070	2	1	12	1
4	50081893237320	101	011	4	1700	2	1	12	1
5	50081893237320	101	011	5	978	2	1	12	1
6	50081893237320	101	011	6	1177	2	1	12	1
7	50081893237320	101	011	7	1102	2	1	12	1
8	50081893237320	101	011	8	1070	2	1	12	1
9	50081893237320	101	011	9	1700	2	1	12	1
10	50081893237320	101	011	10	978	2	1	12	1
11	50081893237320	101	011	11	1102	2	1	12	1
12	50081893237320	101	011	12	1703	2	1	12	1
13	50081893237320	203	011	1	1027	1	.	.	1
14	50081893237320	203	011	2	952	1	.	.	1
15	50081893237320	203	011	3	920	1	.	.	1
16	50081893237320	203	011	4	1550	1	.	.	1
17	50081893237320	203	011	5	828	1	.	.	1
18	50081893237320	203	011	6	1027	1	6	8	1
19	50081893237320	203	011	7	952	1	6	8	1
20	50081893237320	203	011	8	920	1	6	8	1
21	50081893237320	203	011	9	1550	1	.	.	1
22	50081893237320	203	011	10	828	1	.	.	1
23	50081893237320	203	011	11	952	1	.	.	1
24	50081893237320	203	011	12	1553	1	.	.	1
25	50081893237320	102	021	1	1471	2	1	5	2
26	50081893237320	102	021	2	1378	2	1	5	2
27	50081893237320	102	021	3	1338	2	1	5	2
28	50081893237320	102	021	4	2125	2	1	5	2
29	50081893237320	102	021	5	1223	2	1	5	2
30	50081893237320	102	021	6	1471	2	.	.	2
31	50081893237320	102	021	7	1378	2	.	.	2
32	50081893237320	102	021	8	1338	2	.	.	2
33	50081893237320	102	021	9	2125	2	9	12	2
34	50081893237320	102	021	10	1223	2	9	12	2
35	50081893237320	102	021	11	1378	2	9	12	2
36	50081893237320	102	021	12	2128	2	9	12	2

(and so on...)

Working With Survey Weights

SIPP data files include a “person weight” that estimates the number of people in the population each person in the survey represents. Using these weights during analysis is often necessary to create estimates that are representative of the nation. These weights also account for SIPP’s low-income oversample design.

Row	SSUID	PNUM	MONTHCODE	WPFINWGT	(other data...→)
1	10111110312318	101	1	2998.572817	...
2	10111110312318	101	2	2964.092228	...
3	10111110312318	101	3	3024.350114	...
4	10111110312318	101	4	2992.427344	...
5	10111110312318	101	5	2940.215911	...
6	10111110312318	101	6	2993.864952	...
7	10111110312318	101	7	3077.093856	...
8	10111110312318	101	8	3229.440543	...
9	10111110312318	101	9	3283.748056	...
10	10111110312318	101	10	3303.072659	...
11	10111110312318	101	11	3247.880820	...
12	10111110312318	101	12	3237.107828	...
(and so on...)					

The main weighting variable is **WPFINWGT** (“Final person weight”). This variable provides each person’s survey weight for each month of the year. As before, when doing person-level analysis, you’ll likely want to pull one weight per person (usually month 12). This will give you one “annual weight” or “person weight” for each person, such as below.

Row	SSUID	PNUM	WPFINWGT	EDISABL
1	10111110312318	101	3237.107828	2
2	10111110312318	102	1670.366883	1
3	00013364117220	101	2042.482742	1
4	00013364117220	102	2759.639942	2
5	20037909398822	101	1781.053931	2
(and so on...)				

Similarly, when doing household-level analysis, you’ll likely want to pull one weight from the household reference person’s last monthly record (usually month 12). This will give you one “annual weight” or “household weight” for the household, such as below.

Row	SSUID	SHHADID	WPFINWGT	EOWN_VEH
1	10111110312318	011	3237.107828	1
2	00013364117220	011	2042.482742	1
3	20037909398822	011	1781.053931	2
4	20037909398822	021	3317.123814	1
5	00011428546521	011	1972.664142	2
(and so on...)				

Estimating Variances

The Census Bureau also publishes replicate weights to facilitate variance estimation. These weights are available on the “*rw20{XX}*” files and need to be merged with the *Primary Data File* using **SSUID**, **PNUM**, and **MONTHCODE** as merge keys.

Row	SSUID	PNUM	MONTHCODE	WPFINWGT	REPWGT0	REPWGT1	(REPWGT2- REPWGT239)	REPWGT240
1	10111110312318	101	1	2998.572817	2998.572817	2486.960876	...	1204.442091
2	10111110312318	101	2	2964.092228	2964.092228	2471.164682	...	1183.491039
3	10111110312318	101	3	3024.350114	3024.350114	2514.27952	...	1201.00336
4	10111110312318	101	4	2992.427344	2992.427344	2468.824308	...	1182.56291
5	10111110312318	101	5	2940.215911	2940.215911	2490.767414	...	1189.595313
6	10111110312318	101	6	2993.864952	2993.864952	2510.183405	...	1187.36794
7	10111110312318	101	7	3077.093856	3077.093856	2513.102687	...	1175.434143
8	10111110312318	101	8	3229.440543	3229.440543	2526.635926	...	1186.862681
9	10111110312318	101	9	3283.748056	3283.748056	2462.900364	...	1158.439919
10	10111110312318	101	10	3303.072659	3303.072659	2463.35767	...	1152.458048
11	10111110312318	101	11	3247.880820	3247.880820	2459.570474	...	1132.770515
12	10111110312318	101	12	3237.107828	3237.107828	2403.947149	...	1111.151986
(and so on...)								

These weights are all prefixed by “*REPWGT*”. Using our simple form again, you can see this person’s “final person weight” (**WPFINWGT**), as well as their 240 replicate weights. Note that **REPWT0** equals **WPFINWGT**⁵. From here you can perform weighted analyses with variance estimation.

⁵ Respondents only have replicate weights when their **WPFINWGT** is greater than 0.

Contact Us

Have more questions about the SIPP and its data? Please contact us (the *SIPP Coordination and Outreach Staff*) at census.sipp@census.gov. We'd also like to hear what you think of this guide and what other topics you think should be included. All questions and comments are welcome!